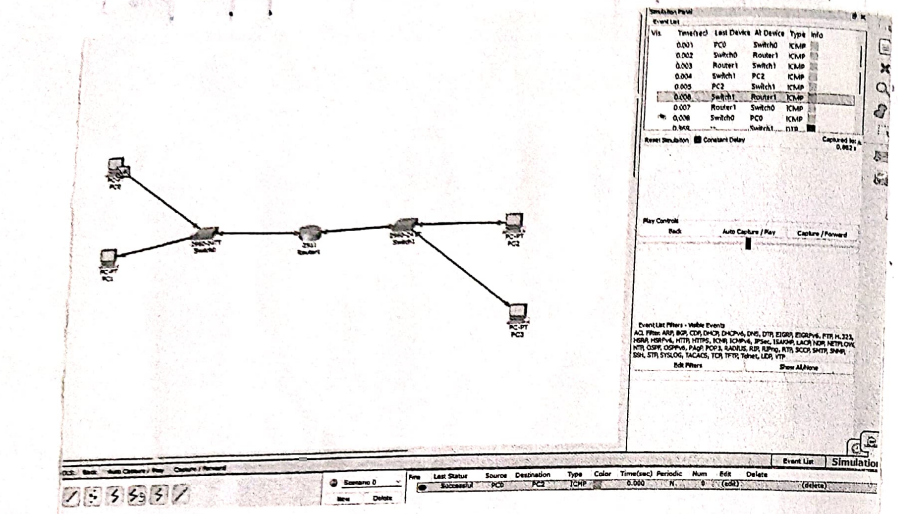
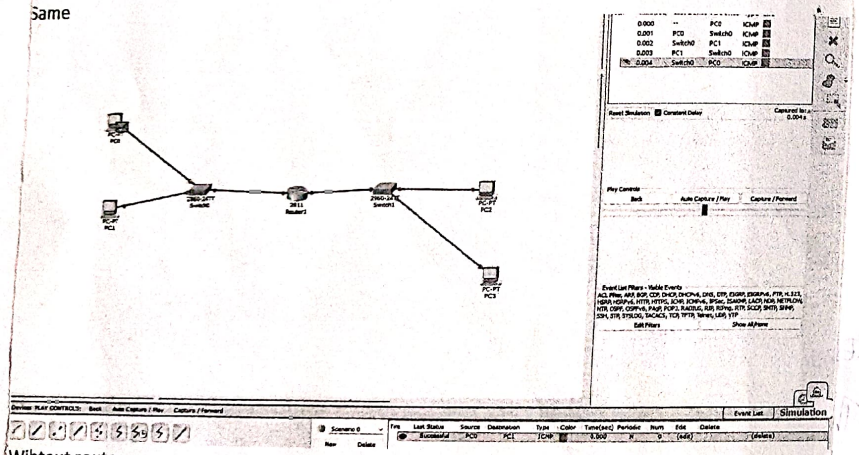
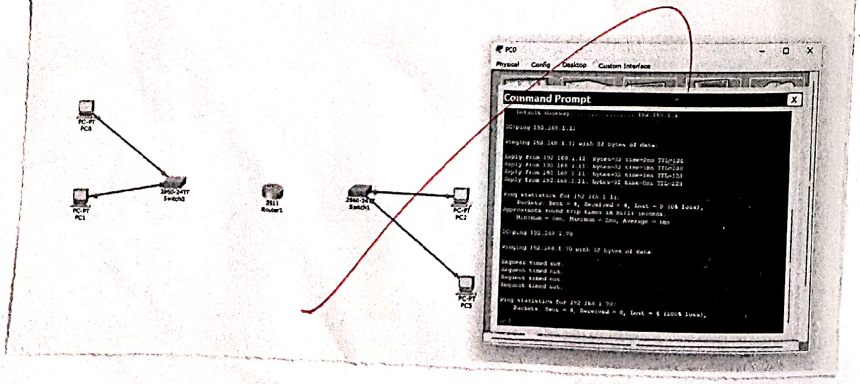


Same



Wihtout router



2018/26

EXPERIMENT 9

IMPLEMENTATION AND SUBNETTING OF CISCO PACKET TRACER

AIM: To design and implement multiple subnets using Cisco Packet Tracer, asign static IP addresses to host connecting subnets, using a router, simulate packet transmission

ALGORITHM:

Step 1: Open Cisco Packet Tracer.

Step 2: Create Multiple LAN / subnets using PC & switch.

Step 3: Select a suitable network address and divide into multiple subnets.

Step 4: Assign static IP address, subnet masks, default gateways.

Step 5: Connect PC's to switch using Cable

Step 6: Connect each switch to user interface

Step 7: Configure the router interface with appropriate IP address.

Step 8: Verify using PING.

Step 9: Use simulation mode to send a packet from 1 PC to another.

Step 10: Observe packet transmission through router.

Step 11: Remove / disconnect router connection.

Step 12: Compare results when subnets are connected

PROCEDURE

1. Create Network Topology:
   - 1 Router - 2911
   - 4 PC
   - 2 Switch - 2960.

2. Create 2 subnets

Subnet 1.

Network Address: 192.168.1.0
Subnet Mask : 255.255.255.192
Usable range : 192.168.1.1 - 192.168.1.62
Broadcast : 192.168.1.63.

Subnet 2:

Network Address: 192.168.1.64
Subnet Mask : 255.255.255.192
Usable Range : 192.168.1.65 - 192.168.1.126
Broadcast : 192.168.1.127

3. Assign IP address to PCs.

4 Configure Router → CLI

enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.192
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.1.65
255.255.255.192

5. Test communication within same subnet.

Run: ping 192.168.1.11.

expected: Reply from 192.168.1.11

Reply from 192.168.1.11.

∴ PC0 and PC1 can communicate within same subnet.

6 Communicate between different subnets.

Run: ping 192.168.1.70

PC0: 192.168.1.0/26

PC2: 192.168.1.64/26

7. Simulate Packet Transmission.

x Add Simple PDU.

PC0 → PC2

PC0 → Switch → Router → Switch1 → PC2

8. Test Without a Router

Now, disconnect the router from switch

Try PC0 → PC2.

or from PC0 → ping 192.168.1.70.

The communication should fail, since the routers are disconnected from switch

RESULT:

Thus the network was successfully subnetted and configured in